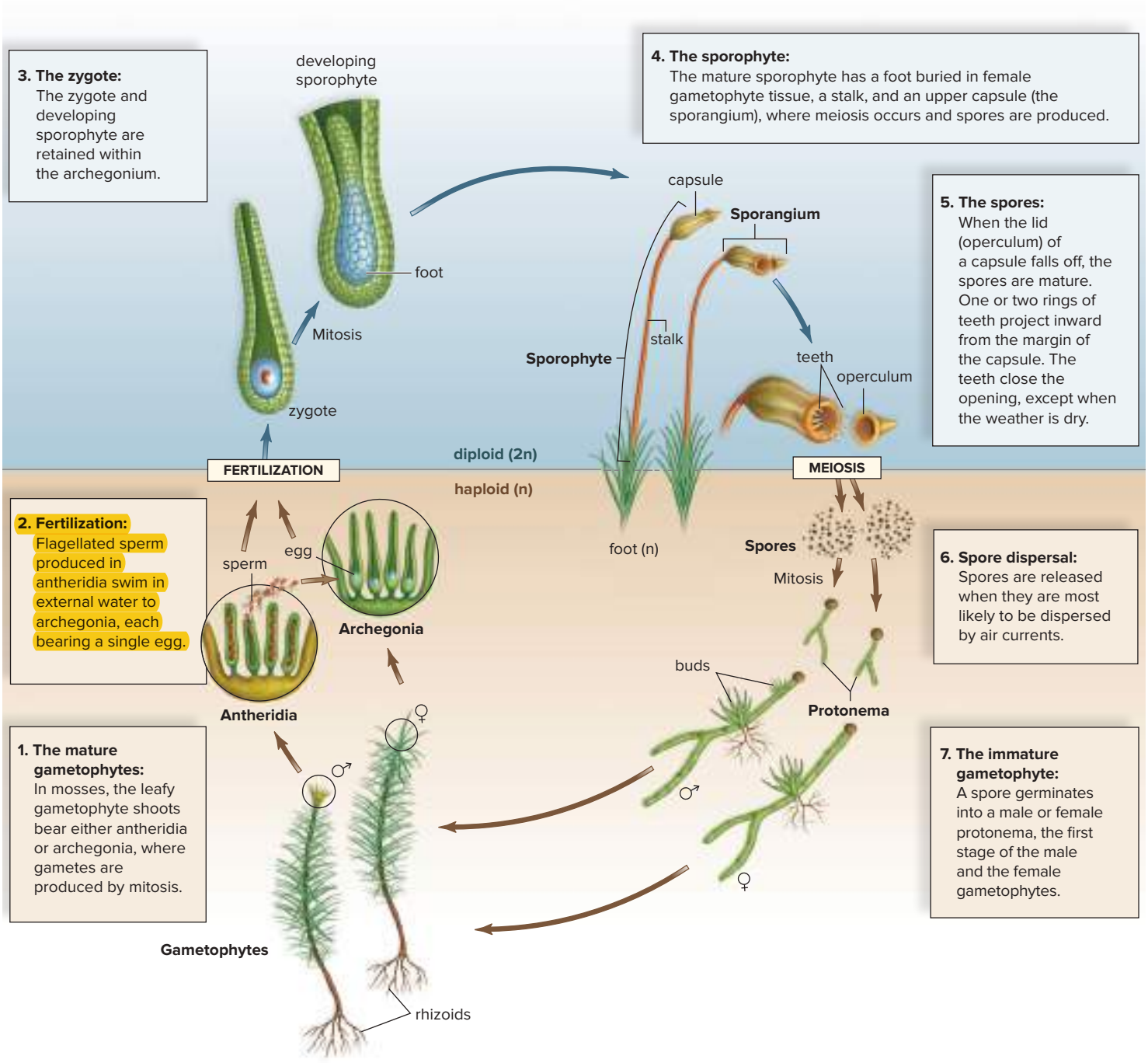


Figure 23.9 Moss life cycle. The majority of a moss life cycle is spent in the haploid (n) state. The sporophyte (2n) is nonphotosynthetic at maturity and produces spores. The leaflike gametophyte (n) is photosynthetic and produces the sperm and egg. The sperm requires water to swim to the egg. This life cycle is from a species of *Polytrichum*.

23.3 Evolution of Lycophytes: Vascular Tissue

Learning Outcomes

Upon completion of this section, you should be able to

1. List the unique structural adaptations found in the lycophytes.
2. Recognize the features present in *Cooksonia* that make it a vascular plant.

Today, **vascular plants,** also known as tracheophytes, dominate the natural landscape in nearly all terrestrial habitats. Vascular plants can achieve great heights because they have roots that absorb water from the soil and a vascular tissue called **xylem** that transports water through the stem to the leaves. Another conducting tissue, called **phloem,** transports nutrients in a plant. Further, the cell walls of the conducting cells in xylem contain **lignin,** a material that strengthens plant cell walls. Thus, the evolution of xylem was essential to the evolution of upright and taller plants.

ISTUDY